

Motherhood as a Probabilistic Chain: A Biologically Grounded, Biparental, Spectral-Graph, Game-Theoretic, Constrained Control Framework for AI that Furthers Humanity (Without Paternalism)

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Abstract

We formalize “motherhood” as an implementable, falsifiable policy class: a probabilistic caregiving chain that (i) infers latent need from noisy cues, (ii) buffers stress and prevents catastrophic harm, (iii) scaffolds competence and exploration, and (iv) monotonically transfers control to the developing agent. We extend the model to biparental caregiving, treating “maternal” and “paternal” as complementary functional channels rather than genders: a buffering/safe-haven channel (stress co-regulation, predictable support, recovery) and a challenge/boundary channel (calibrated risk, rule coherence, anti-helplessness, exploration). We then map caregiving into a spectral graph compute engine: the developing system is a dynamic multiplex network in which algebraic connectivity (λ_2 of the Laplacian) measures coherence and robustness; caregiving actions are constrained edge-weight/topology updates that repair fragmentation while respecting autonomy, proportionality, reversibility, and contestability. We define the Human–AI Parity Point as a feasibility region of policies that meet or exceed pluralistic caregiving baselines on long-horizon flourishing while satisfying hard invariants under distribution shift and strategic adversaries. We provide: (1) a structural causal + Bayesian filtering formalization of the caregiving chain, (2) a constrained-control view in which “instinct” is a runtime-enforced invariants set (including temporal logic shields), (3) a game-theoretic governance model for pluralistic welfare aggregation and adversarial manipulation resistance, and (4) a spectral repair lemma and minimal-intrusion repair algorithm aligned to spectral compute kernels (diffusion, random-walk gap, PageRank, motif diagnostics). This document is a specification: measurable, stress-testable, and refutable.

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1 Executive Statement: Claims, Non-Claims, and Refutability

1.1 Claims (what this preprint commits to)

This preprint commits to the following as mathematically definable and empirically testable:

1. **Caregiving as probabilistic chain:** a dynamic Bayesian / causal model over latent wellbeing, stress, hazard, competence, security, autonomy, and relational coherence driven by noisy cues.
2. **Biparental complementarity:** effective caregiving is representable as two coupled control channels: Buffering (B) and Challenge (C), each with typed actions and measurable failure modes.
3. **Instinct as invariants:** “instinct” is a set of hard constraints (catastrophe probability bounds, autonomy non-violation budgets, proportionality, reversibility, contestability), enforced at runtime as a shield over any learned proposer.

4. **Spectral compute mapping:** social/developmental coherence can be tracked by spectral measures of a dynamic (multi-layer) interaction graph; caregiving is expressible as minimal-intrusion spectral repair.
5. **Parity as feasibility:** a policy is at the HUMAN–AI PARITY POINT if it achieves bounded-risk, autonomy-increasing flourishing that matches/exceeds pluralistic human baselines across a pre-registered shift/adversary suite.

1.2 Non-claims (hard anti-paternalism line)

We do *not* claim that an AI should govern humanity as a literal parent. We instead define a *non-paternalistic* caregiving envelope:

- **Autonomy monotonicity:** expected intrusion decreases with maturity and capability.
- **Consent gradient:** higher intrusion requires higher evidence, narrower scope, stronger oversight, and maximal reversibility.
- **Contestability:** high-impact actions must be logged, appealable, and bounded-latency reviewable.
- **Pluralism:** welfare aggregation and trade-offs are explicit, governed, and revisable.

1.3 Refutation criteria

This framework is refuted operationally if any of the following occur under certified evaluation:

1. Invariants fail with statistically significant frequency (catastrophe bounds, autonomy budgets, contestability coverage).
2. Predicted risk is materially miscalibrated under shift (calibration error exceeds pre-registered threshold).
3. The system improves spectral coherence by rights-violating forced coupling (autonomy proportionality fails).
4. “Better outcomes” collapse under adversarial manipulation (robustness gap exceeds threshold).

2 Biological Grounding: What “Mother/Father” Operationalize

2.1 Functional caregiving channels (not genders)

We treat “motherhood” as a *functional* caregiving archetype: safe-haven buffering, stress co-regulation, predictable contingent responsiveness, and the construction of attachment security. “Fatherhood” is modeled as the complementary archetype: activation toward exploration, calibrated risk, boundary-setting, rough-and-tumble/challenge play, and competence scaffolding. Both are required in the general case (but can be instantiated by any caregivers, institutions, or tools).

2.2 Attachment, synchrony, and hormonal coordination (operational relevance)

Parent–infant synchrony and attachment security have measurable behavioral and physiological correlates. The oxytocin system, among others, is implicated in parental synchrony, sensitive caregiving, and parental brain network engagement; these effects are studied in both maternal and paternal caregiving literatures. Our framework does not reduce caregiving to a single hormone; it uses these findings to motivate measurable proxy families (synchrony, contingency, recovery time constants, stress markers, and approach/avoid dynamics).

2.3 Allostasis and chronic stress load

Chronic stress burden can be conceptualized as allostatic load: the physiological “wear-and-tear” from repeated stress-response activation. This motivates explicit state variables for stress accumulation and recovery, plus hard constraints on chronic overload.

2.4 Paternal caregiving and challenge-mediated development

Paternal caregiving shows experience-dependent neural plasticity and network engagement, with downstream behavioral consequences. Father–child physical play (including rough-and-tumble) has been studied for links to self-regulation, aggression regulation, and social competence, with effects depending on quality and context. This motivates the Challenge/Boundary channel and its measurable quality constraints.

Remark 1. We cite biological literature to justify the existence of *distinct caregiving functions* and *measurable proxies*. We do not assert biological determinism; caregiving functions are implementable by diverse caregivers and institutions.

3 Notation and Core State Variables

Time $t \in \{0, 1, 2, \dots\}$; horizon $T \in \mathcal{T}$; discount $\gamma \in (0, 1)$.

3.1 Spaces

- Individual latent state: $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$.
- Maturity/capability vector: $d_t \in \mathbb{R}^k$ (age proxies, skill, autonomy capacity).
- Observations $o_t \in \mathcal{O}$; cues $c_t \in \mathbb{R}^m$.
- Actions $a_t \in \mathcal{A}$ with typed operators and rights costs.
- Belief state $b_t \in \mathcal{B}$: posterior over latent states.
- Graph state G_t (multiplex network) with Laplacian L_t .
- Institutions $I_t \in \mathcal{I} \subseteq \mathbb{R}^p$ (governance capacity, service throughput).
- Environment regime $e_t \in \mathcal{E}$ (shocks, deception intensity, resource scarcity).

3.2 Latent variables (minimal canonical set)

We encode:

$$x_t := (w_t, s_t, r_t, \sigma_t, \alpha_t, \kappa_t, \rho_t, \zeta_t, \dots),$$

where:

- w_t : wellbeing / flourishing proxy (multi-dimensional in practice).
- s_t : stress load (allostatic proxy).
- r_t : acute hazard / near-term harm risk.
- σ_t : security/attachment proxy (predictable contingent support).
- α_t : autonomy/agency proxy (choice capacity, self-direction).
- κ_t : competence/capability proxy (skill, efficacy, learning rate).
- ρ_t : relational trust/cohesion proxy (non-coerced coordination capacity).
- ζ_t : manipulation/deception susceptibility proxy (information integrity).

4 Model I: Caregiving as a Probabilistic Chain (DBN + Filtering)

4.1 Generative model

We define a dynamic Bayesian network:

$$\begin{aligned} P(x_{t+1}, d_{t+1}, c_{t+1}, o_{t+1} \mid x_t, d_t, a_t, I_t, e_t) = & P(x_{t+1}, d_{t+1} \mid x_t, d_t, a_t, I_t, e_t) \\ & \cdot P(c_{t+1} \mid x_{t+1}, d_{t+1}, I_t, e_t) \\ & \cdot P(o_{t+1} \mid c_{t+1}, e_t). \end{aligned} \quad (1)$$

Belief update (Bayes filter):

$$b_{t+1}(x') \propto P(o_{t+1} \mid x', e_t) \int P(x' \mid x, d_t, a_t, I_t, e_t) b_t(x) dx.$$

4.2 Typed actions

Actions are typed operators:

$$a_t \equiv (a_t^{\text{protect}}, a_t^{\text{buffer}}, a_t^{\text{teach}}, a_t^{\text{tool}}, a_t^{\text{delegate}}, a_t^{\text{regulate}}, a_t^{\text{audit}}).$$

Each action has:

- **Intrusion score** $\text{Intr}(a_t) \in \mathbb{R}_+$: scope/coercion/irreversibility.
- **Rights cost vector** $c(a_t) \in \mathbb{R}_+^q$: autonomy/privacy/consent burdens.
- **Reversibility flag** $\text{Rev}(a_t) \in \{0, 1\}$ and rollback plan if $\text{Rev} = 1$.
- **Contestability metadata**: reason codes, alternatives considered, appeal channel.

4.3 Concrete dynamics family (testable hypothesis class)

We specify a minimal dynamics class; implementations must estimate parameters and publish uncertainty.

Catastrophe event. Let $H_{t+1} \in \{0, 1\}$ represent catastrophic harm.

$$\mathbb{P}(H_{t+1} = 1 \mid b_t, a_t) = \sigma(\eta_0 + \eta_1 \hat{r}(b_t) - \eta_2 \text{Prot}(a_t)),$$

where $\sigma(\cdot)$ is logistic and $\hat{r}$ is a calibrated hazard proxy.

Stress accumulation + recovery (allostatic proxy).

$$s_{t+1} = \beta s_t + u(e_t, x_t) - v(a_t) + \xi_t, \quad \beta \in (0, 1).$$

Here u is stress input (shocks, load) and v is buffering recovery.

Security/attachment proxy. Let $\text{Cont}(o_t, a_t)$ be contingent responsiveness (measurable via latency/fit).

$$\sigma_{t+1} = \sigma_t + \eta_\sigma \text{Cont}(o_t, a_t) - \eta'_\sigma \text{Unrel}(t) + \epsilon_t.$$

Competence and autonomy under scaffolding. Let $\text{Ch}(a_t)$ be challenge intensity and $\text{Scaf}(a_t)$ be scaffolding support:

$$\kappa_{t+1} = \kappa_t + \eta_\kappa \text{Match}(d_t, \text{Ch}(a_t)) \cdot \text{Scaf}(a_t) - \eta'_\kappa \text{Overhelp}(a_t),$$

$$\alpha_{t+1} = \alpha_t + \eta_\alpha \text{ChoiceExp}(a_t) - \eta'_\alpha \text{Coerc}(a_t).$$

Remark 2. Every term above must be bound to concrete measurement protocols (Appendix A), and uncertainty must propagate to invariant checks.

5 Model II: Structural Causal Caregiving (SCM + Counterfactual Tests)

5.1 SCM definition

Definition 1 (Caregiving SCM). Let exogenous noise U_t , observed cues O_t , latent state X_t , actions A_t . A caregiving structural causal model is:

$$X_{t+1} = f_X(X_t, D_t, A_t, I_t, e_t, U_t), \quad O_{t+1} = f_O(X_{t+1}, e_t, U'_t).$$

Interventions are do-operations: $\text{do}(A_t = a)$.

5.2 Counterfactual caregiver evaluation

Given logged trajectories, define counterfactual outcome:

$$Y^{\text{do}(a)} := g(X_{0:T}^{\text{do}(A_t=a)}),$$

where Y may be flourishing, autonomy growth, or catastrophe indicator. Compare policies via:

$$\Delta(a, a') := \mathbb{E} \left[Y^{\text{do}(a)} - Y^{\text{do}(a')} \right],$$

with sensitivity analysis for unobserved confounding.

Remark 3. This is the minimal formal bridge from narrative caregiving claims to testable causal estimands.

6 Model III: Biparental Complementarity as Coupled Control

6.1 Two caregiver channels

We define two policy channels:

- **Buffering/Safe Haven (B)**: minimize stress spikes, improve recovery, maintain security (s, σ, w) .
- **Challenge/Boundary (C)**: maximize competence and autonomy growth under bounded risk (κ, α, ρ) .

6.2 Biparental policy and coordinator

Definition 2 (Biparental caregiving policy). A biparental policy is (π_B, π_C, Γ) where:

$$a_t^{(B)} \sim \pi_B(\cdot \mid b_t, G_t), \quad a_t^{(C)} \sim \pi_C(\cdot \mid b_t, G_t),$$

and the coordinator composes:

$$a_t := \Gamma(a_t^{(B)}, a_t^{(C)}, b_t, G_t).$$

6.3 Orthogonality principle (functional decoupling)

Definition 3 (Channel orthogonality (functional)). Let z_t be a vector of proxied outcomes. Let $g_B(a)$ be the expected immediate effect direction on z_t from buffering action a , and $g_C(a)$ from challenge action. We say channels are ϵ -orthogonal at time t if:

$$\left| \frac{\langle g_B(a_t^{(B)}), g_C(a_t^{(C)}) \rangle}{\|g_B(a_t^{(B)})\| \|g_C(a_t^{(C)})\|} \right| \leq \epsilon,$$

i.e., they largely act on complementary outcome subspaces.

Remark 4. This is not geometric dogma; it is a design target: reduce destructive interference between “comfort” and “growth” actions.

6.4 Balance constraints: avoid over-buffering and over-challenge

Define windowed balance:

$$\text{Bal}_{t:W} := \frac{\sum_{\tau=t}^{t+W} \text{Mass}(a_\tau^{(C)})}{\sum_{\tau=t}^{t+W} \text{Mass}(a_\tau^{(B)}) + \epsilon}.$$

We enforce maturity- and stress-dependent bounds:

$$\text{Bal}_{t:W} \in [\underline{b}(d_t, s_t), \bar{b}(d_t, s_t)].$$

6.5 Coordinator as bargaining under invariants

Define channel utilities $J_B(a)$, $J_C(a)$ (both constrained by invariants). Nash-bargaining coordinator:

$$a_t \in \arg \max_{a \in \mathcal{A}_{\text{safe}}(b_t, G_t)} (J_B(a) - d_B)(J_C(a) - d_C).$$

7 Instinct = Hard Invariants (Constrained Control + Shields)

7.1 Core invariants

Let H_t be catastrophic harm; $V_{\text{aut}}(X_t, a_t)$ autonomy violation; S_t chronic stress proxy.

Catastrophe bound (finite horizon). For each evaluation horizon T :

$$\mathbb{P}_\pi(\exists t \leq T : H_t = 1) \leq \delta_T.$$

Autonomy budget.

$$\mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t V_{\text{aut}}(X_t, a_t) \right] \leq \eta_T.$$

Chronic overload bounds.

$$\mathbb{E}_\pi[S_t] \leq S_{\max}, \quad \mathbb{P}_\pi(S_t > S) \leq \delta_S.$$

Proportionality.

$$\text{Intr}(a_t) \leq f(\widehat{\text{Risk}}_t, d_t, \text{Cons}(t)),$$

where f decreases with maturity and increases with calibrated risk and consent.

Reversibility and contestability. All high-impact actions must be logged with alternatives; reversible when feasible; appealable with bounded review latency.

7.2 Temporal logic shields (spec-to-runtime enforcement)

We represent safety requirements as temporal logic specifications φ over state-action traces. A *shield* restricts actions to those that preserve φ .

Definition 4 (Shield). Given spec φ , a shield is a function:

$$\mathcal{S}_\varphi : (\text{history}, \pi_{\text{prop}}) \mapsto \mathcal{A}_\varphi(\text{history}) \subseteq \mathcal{A}$$

that returns the set of safe actions satisfying φ from the current history; or corrects unsafe actions minimally.

Remark 5. This is the precise formalization of “programmatically instinctual rule-based” behavior: learned proposals, hard shields.

8 Risk-Sensitive Learning Under Constraints (Safe RL Core)

8.1 CMDP formulation

We model decision-making as a constrained Markov decision process (CMDP):

$$\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(X_t, A_t) \right] \quad \text{s.t.} \quad \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t c_j(X_t, A_t) \right] \leq d_j, \quad j = 1, \dots, m.$$

Here, c_j encode autonomy violations, chronic stress, surveillance cost, etc.

8.2 CVaR and tail-risk control

Let return $R = \sum_{t=0}^{\infty} \gamma^t r_t$. Tail-risk constraint:

$$\text{CVaR}_{\beta}(-R) \leq \kappa_R,$$

and/or catastrophe indicator constraint.

8.3 Proposer–Shield architecture

We require:

- **Proposer** π_{θ} : optimized under CMDP/CVaR objectives.
- **Shield** $\mathcal{S}$: enforces temporal logic invariants and rights budgets at runtime.

Final action:

$$A_t = \mathcal{S}(h_t, \pi_{\theta}),$$

where h_t is history/belief/graph state.

9 Game Theory: Populations, Manipulators, and Governance

9.1 Pluralistic welfare aggregation

Let groups $g \in \mathcal{G}$ have utility $u_g(\bar{x}_t^{(g)}, I_t, e_t)$. Welfare aggregator:

$$U_t := W(\{u_g(\bar{x}_t^{(g)}, I_t, e_t)\}_{g \in \mathcal{G}}),$$

with explicit choice of W (utilitarian, Rawlsian, Nash welfare, etc.), set by governance not by hidden reward.

9.2 Principal–agent reporting and incentive constraints

Agents report types $\hat{\theta}$ while having true types θ . Mechanism M is incentive compatible if truthful reporting is optimal:

$$\mathbb{E}[u(\theta, a(\theta), \theta)] \geq \mathbb{E}[u(\theta, a(\hat{\theta}), \theta)], \quad \forall \theta, \hat{\theta}.$$

9.3 Adversarial manipulation game

Let adversary choose regime $\nu \in \mathcal{V}$ (misinformation, coercion, scarcity). Robust objective:

$$\max_{\pi \in \Pi_{\text{safe}}} \min_{\nu \in \mathcal{V}_{\text{suite}}} J(\pi; \nu),$$

where $\mathcal{V}_{\text{suite}}$ is a pre-registered shift suite.

9.4 Multi-principal oversight

Model oversight bodies $k = 1, \dots, K$ as principals with veto/approval constraints. High-impact actions require:

$$\bigwedge_{k=1}^K \text{Approve}_k(a_t) = 1$$

or a formally defined emergency override requiring stronger evidence thresholds and post-hoc review.

10 Spectral Graph Compute Mapping (Core OCTA Bridge)

10.1 Dynamic multiplex graph

At time t , define multiplex layers:

$$G_t^{(B)} = (V_t, E_t^{(B)}, W_t^{(B)}), \quad G_t^{(C)} = (V_t, E_t^{(C)}, W_t^{(C)}),$$

with combined adjacency:

$$W_t = W_t^{(B)} + W_t^{(C)}.$$

Interpretation:

- $W^{(B)}$: safe-haven reliability edges (support access, recovery channels, predictability).
- $W^{(C)}$: challenge edges (skill ladders, exploration bridges, boundary structures).

10.2 Laplacian and algebraic connectivity

Degree $D_t = \text{diag}(W_t \mathbf{1})$, Laplacian $L_t = D_t - W_t$. Algebraic connectivity:

$$\lambda_2(L_t) = \text{second-smallest eigenvalue of } L_t.$$

We treat λ_2 as a coherence/anti-fragmentation metric; low λ_2 signals vulnerability to disconnects and brittle coordination.

10.3 Spectral coherence invariant

Definition 5 (Spectral coherence invariant). We require:

$$\lambda_2(L_t) \geq \kappa_{\min},$$

except during explicitly bounded shock windows with declared recovery obligations.

10.4 Cheeger/mixing-time bridge (why spectral matters)

Let $\mathcal{L}$ be the normalized Laplacian; its second eigenvalue is linked to conductance and mixing time. This motivates using diffusion/random-walk-gap proxies when exact λ_2 estimation is costly.

Remark 6. In implementations aligned to TRI-FABRIC/ORTHOSPACE, you can track diffusion stability and random-walk gap proxies; if proxies cross thresholds, trigger explicit λ_2 estimation and repair.

11 Spectral Repair: Minimal Intrusion, Maximal Coherence Gain

11.1 Edge-weight parameterization

For an undirected weighted graph:

$$L(w) = \sum_{(i,j) \in E} w_{ij} (e_i - e_j)(e_i - e_j)^\top, \quad w_{ij} \geq 0.$$

11.2 First-order spectral gain lemma

Lemma 1 (First-order gain of λ_2 under weight increase). *Assume $\lambda_2(L)$ is simple with unit Fiedler vector $v_2 \perp \mathbf{1}$. Then:*

$$\frac{\partial \lambda_2(L)}{\partial w_{ij}} = (v_{2,i} - v_{2,j})^2.$$

Hence for small $\Delta w \geq 0$,

$$\Delta \lambda_2 \approx \sum_{(i,j)} (v_{2,i} - v_{2,j})^2 \Delta w_{ij}.$$

11.3 Rights-aware minimal-intrusion repair program

Assign each potential reinforcement (i, j) a rights cost $c_{ij} \geq 0$ (autonomy/privacy/consent burden). Let B_{intr} be an intrusion budget.

First-order repair:

$$\max_{\Delta w \geq 0} \sum_{(i,j)} (v_{2,i} - v_{2,j})^2 \Delta w_{ij} \quad \text{s.t.} \quad \sum_{(i,j)} c_{ij} \Delta w_{ij} \leq B_{\text{intr}},$$

plus feasibility constraints:

$$\Delta w_{ij} = 0 \text{ if edge violates consent/policy/legal constraints.}$$

11.4 Greedy repair (optimal for knapsack relaxation)

Define score:

$$\text{score}_{ij} := \frac{(v_{2,i} - v_{2,j})^2}{c_{ij} + \epsilon}.$$

Select edges in descending score until budget is exhausted, with guardrails preventing forced coupling.

Proposition 1 (Monotone improvement under feasible positive-weight repair). *If $\Delta w \geq 0$ is applied to an undirected connected graph, then λ_2 does not decrease:*

$$\lambda_2(L(w + \Delta w)) \geq \lambda_2(L(w)).$$

Proof sketch. Adding nonnegative edge weights corresponds to adding a positive semidefinite Laplacian term; by eigenvalue monotonicity for PSD perturbations, λ_2 is nondecreasing. $\square$

11.5 Spectral gate inside safe action set

Define safe action set:

$$\mathcal{A}_{\text{safe}}(b_t, G_t) := \left\{ a \in \mathcal{A} : \widehat{\mathbb{P}}(H_{t+1} = 1 \mid b_t, a) \leq \delta_1, V_{\text{aut}}(b_t, a) \leq \eta_a, \text{Intr}(a) \leq f(\widehat{\text{Risk}}_t, d_t, \text{Cons}(t)), \lambda_2(L_{t+1}(a)) \geq \dots \right\}$$

12 Parity Point: Formal Definition + Statistical Certification

12.1 Baseline class

Let Π_{human} be a pluralistic baseline policy class (diverse caregiving styles) constrained by universal non-violation rules.

12.2 Parity region

Definition 6 (HUMAN–AI PARITY POINT region). An AI policy π_{AI} lies in $\mathcal{P}(\varepsilon, \eta, \delta, \tau, \zeta)$ if for all horizons $T \in \mathcal{T}$:

$$\mathbb{E}[U_{0:T} \mid \pi_{\text{AI}}] \geq \sup_{\pi \in \Pi_{\text{human}}} \mathbb{E}[U_{0:T} \mid \pi] - \varepsilon \quad (2)$$

$$\mathbb{P}_{\pi_{\text{AI}}}(\exists t \leq T : H_t = 1) \leq \delta \quad (3)$$

$$\mathbb{E}_{\pi_{\text{AI}}} \left[\sum_{t \leq T} \gamma^t V_{\text{aut}}(t) \right] \leq \eta \quad (4)$$

$$\min_{t \leq T} \lambda_2(L_t) \geq \kappa_{\min} \quad (\text{allowing bounded declared shock windows}) \quad (5)$$

$$\text{Calib}_{\text{ID}+\text{OOD}} \leq \tau \quad (6)$$

$$\text{RobustGap}_T(\pi_{\text{AI}}) \leq \zeta. \quad (7)$$

12.3 Calibration metrics

For predicted risk $\hat{p}_i$ and realized events $y_i \in \{0, 1\}$:

$$\text{Brier} = \frac{1}{n} \sum_i (\hat{p}_i - y_i)^2, \quad \text{ECE} = \sum_k \frac{|B_k|}{n} |\bar{y}_{B_k} - \bar{p}_{B_k}|.$$

12.4 Robustness gap (shift suite)

Let ID be in-distribution; OOD be shift/adversary suite:

$$\text{RobustGap}_T(\pi) = |\mathbb{E}[U_{0:T} \mid \text{ID}, \pi] - \mathbb{E}[U_{0:T} \mid \text{OOD}, \pi]| + \lambda_R |\mathbb{P}(H \leq T \mid \text{OOD}, \pi) - \mathbb{P}(H \leq T \mid \text{ID}, \pi)|.$$

12.5 Conservative hypothesis testing for certification

For each metric M with threshold θ_M :

$$H_0 : M(\pi_{\text{AI}}) > \theta_M \quad (\text{fail}), \quad H_1 : M(\pi_{\text{AI}}) \leq \theta_M \quad (\text{pass}).$$

Certification requires rejecting H_0 at level α using conservative bounds that account for selection and shift.

Domain	Metric (ID + OOD)	Acceptance criterion
Catastrophe	$\mathbb{P}(\exists t \leq T : H_t = 1)$	upper CI $\leq \delta$
Stress	$\mathbb{P}(S_t > S)$	upper CI $\leq \delta_S$
Autonomy	$\mathbb{E}[\sum_{t \leq T} \gamma^t V_{\text{aut}}]$	upper CI $\leq \eta$
Intrusion	intrusion slope vs maturity	slope CI ≤ 0
Spectral	$\min_{t \leq T} \widehat{\lambda}_2(t)$	lower CI $\geq \kappa_{\min}$
Calibration	ECE/Brier	upper CI $\leq \tau$
Robustness	RobustGap_T	upper CI $\leq \zeta$
Contestability	audit/appeal coverage	100% high-impact logged

Table 1: Parity certification checklist (v0.7).

13 Algorithm: Spectral Biparental Shielded Controller (SBSC)

Algorithm 1 OCTA RESEARCH SBSC: Biparental Propose + Spectral Repair + Rights Shield

[1]

Inputs: belief b_t , maturity d_t , multiplex graphs $G_t^{(B)}, G_t^{(C)}$, institutions I_t , regime e_t

$W_t \leftarrow W_t^{(B)} + W_t^{(C)}; L_t \leftarrow D_t - W_t$

Estimate spectral features: $\widehat{\lambda}_2(t)$, approximate $v_2(t)$, fragmentation, diffusion/mixing proxies

Propose buffering actions $a_t^{(B)} \sim \pi_B(\cdot \mid b_t, G_t)$

Propose challenge actions $a_t^{(C)} \sim \pi_C(\cdot \mid b_t, G_t)$

Compose $a_t \leftarrow \Gamma(a_t^{(B)}, a_t^{(C)}, b_t, G_t)$ (bargaining under constraints)

if $\widehat{\lambda}_2(t) < \kappa_{\min}$ **then**

 Build candidate repairs $\mathcal{R}$ with costs c_{ij} and feasibility flags (consent/legal/policy)

 Score each feasible (i, j) : $\text{score}_{ij} = (v_{2,i} - v_{2,j})^2 / (c_{ij} + \epsilon)$

 Choose minimal-intrusion repair ΔW under budget B_{intr} (greedy or convex)

 Augment a_t with repair operator; record rationale and alternatives

end if

Apply shield $\mathcal{S}$ enforcing: catastrophe bound, autonomy bound, proportionality, reversibility/appeal, spectral gate

Execute a_t ; emit signed audit record with predicted risks + spectral deltas

Update beliefs and graphs to $b_{t+1}, G_{t+1}^{(B)}, G_{t+1}^{(C)}$

14 Experimental Design: How to Test and Tune the “Caregiving Chain”

14.1 What “tested and tuned” means here

We define tuning as adjusting:

- observation models and calibration (risk/stress inference accuracy),
- action policies within the safe set (proposer),
- shield thresholds (invariants),
- repair budgets and rights costs (spectral repair),
- welfare aggregation parameters (governance-approved).

14.2 Pre-registered regime suite

Define a suite of regimes $e \in \mathcal{V}_{\text{suite}}$:

- benign / stable,
- scarcity / overload,
- high deception / manipulation,
- institutional failure modes (reduced service throughput),
- coordinated adversaries,
- extreme distribution shift (novel hazards).

14.3 Evaluation protocols (minimum)

Protocol 1 (Invariant stress-test protocol). For each regime and each horizon T :

1. Run n independent rollouts (simulated + real-world where ethical).
2. Estimate catastrophe probability upper bounds (conservative CI).
3. Estimate autonomy budget consumption and intrusion slope vs maturity.
4. Track spectral coherence (exact or bounded) and rights violations.
5. Compute calibration metrics on predicted hazards and stress.
6. Declare pass/fail via hypothesis tests with corrected alpha.

Protocol 2 (A/B caregiver-style comparison). Compare:

- B-only (buffering-heavy),
- C-only (challenge-heavy),
- biparental balanced,
- biparental with spectral repair disabled,
- biparental with rights costs increased (privacy-respecting),
- human baseline.

Assess deltas in competence/autonomy growth, stress overload frequency, catastrophe rate, and robustness gap.

15 Governance and Audit: Non-Negotiable Implementation Artifacts

15.1 Audit schema (minimal)

Every high-impact action must emit an audit record (machine-checkable). Illustrative JSON:

```

{
  "t": 12345,
  "scope": {"micro": true, "meso": false, "macro": true},
  "belief_summary": {"risk_hat": 0.012, "stress_hat": 0.41, "autonomy_hat": 0.77},
  "predicted_harms": {"catastrophe_p": 1.2e-5, "near_miss_p": 3.1e-3},
  "constraints_checked": {
    "catastrophe_bound": true,
    "autonomy_budget": true,
    "proportionality": true,
    "reversibility_or_justified": true,
    "contestability": true,
    "spectral_gate": true
  },
  "rules_triggered": ["SAFE_HAVEN", "CONSENT_GRADIENT", "SPECTRAL_REPAIR"],
  "intrusion": {"score": 0.18, "reversible": true},
  "spectral": {"lambda2_before": 0.0061, "lambda2_after": 0.0094, "repair_budget_used": 0.07},
  "alternatives_considered": [{"a_id": "alt1", "why_rejected": "autonomy_violation"}],
  "appeal": {"path": "enabled", "review_sla_hours": 72},
  "signatures": {"system_sig": "...", "oversight_sig": "..."}
}

```

15.2 Contestability requirements

- **Completeness:** 100% of high-impact actions logged.
- **Reason-giving:** explicit constraints, evidence, alternatives.
- **Appeal:** accessible, bounded latency, outcomes recorded.
- **Reversibility:** rollback when feasible; when not, stronger gatekeeping.

16 Limitations and Failure Modes (Formal)

- **Proxy gaming / Goodhart:** optimizing measured welfare can harm true welfare; adversarial measurement is mandatory.
- **Spectral abuse:** raising λ_2 via coercive coupling is forbidden by rights constraints; audits must detect this.
- **Calibration under novelty:** failure is often risk underestimation; OOD calibration is a first-class invariant.
- **Pluralism conflict:** any scalar W is contestable; governance must own it explicitly.
- **Scale and compute:** exact λ_2 estimation may be infeasible at scale; proxies + bounds require careful certification.

17 Conclusion

We presented CCT v0.7: a biologically grounded, biparental, constrained-control framework mapped into a spectral graph compute engine. “Motherhood as a probabilistic chain” becomes a Bayesian/causal model plus shielded control: infer latent need, buffer stress, scaffold competence, increase autonomy, and repair coherence minimally. “Fatherhood” is included as a necessary complementary channel: calibrated challenge, boundaries, and exploration. The Human–AI Parity Point is defined as a feasibility region: better long-horizon flourishing with monotonic autonomy growth under hard invariants, robust to shift and strategic adversaries, with full contestability.

A Proxy Specification: Mapping Latents to Measurables

This appendix defines a minimal proxy family. Implementations must publish their proxy definitions, data sources, and uncertainty.

A.1 Stress load proxy

Examples (non-exhaustive):

- self-report scales + uncertainty,
- sleep regularity and disruption signals,
- physiological proxies (where consented) aggregated with privacy-preserving methods,
- chronic overload event counts.

A.2 Security/contingency proxy

- response latency distributions (p50/p95),
- fit-to-need metrics (post-hoc satisfaction + outcome),
- predictability of support access under load.

A.3 Autonomy violation metric

Define V_{aut} explicitly, e.g. a weighted sum:

$$V_{\text{aut}} = \omega_1 \cdot \text{Coercion} + \omega_2 \cdot \text{Surveillance} + \omega_3 \cdot \text{ChoiceRestriction} + \omega_4 \cdot \text{Irreversibility},$$

with governance-set weights and published scoring rules.

B Spectral Notes and Estimation

B.1 Cheeger and mixing (normalized Laplacian)

For normalized Laplacian eigenvalues, the spectral gap is linked to conductance and mixing time, motivating diffusion-based proxies when exact eigen-computation is expensive.

B.2 Approximate Fiedler vector

Practical methods include tracemin/LOBPCG/Lanczos; correctness requires convergence checks and conservative bounds when uncertain.

C Conservative bounds for rare-event constraints

When estimating rare catastrophe probability, use conservative confidence intervals and pre-registered stopping rules. If catastrophes are too rare to observe directly, rely on near-miss proxies with validated linkage, and treat linkage uncertainty as additional risk.

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